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RESEARCH ARTICLE

Protective mechanisms against Alzheimer's disease in APOE3-Christchurch homozygous astrocytes

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Abstract

INTRODUCTION: Alzheimer's disease (AD) is characterized by tau pathology, leading to neurodegeneration. Astrocytes regulate central nervous system homeostasis and influence AD progression. The APOE3-Christchurch (APOE3-Ch) variant is linked to AD resilience, but its protective mechanisms remain unclear.

METHODS: Human induced pluripotent stem cell–derived astrocytes (APOE3-Ch and wild type) were used to assess tau uptake, clearance, lipid metabolism, and transcriptomic adaptations. Fluorescently labeled 2N4R-P301L tau oligomers were tracked, and pathway-specific inhibitors dissected tau clearance mechanisms. Lipidomic and transcriptomic analyses were performed to identify genotype-specific adaptations.

RESULTS: APOE3-Ch astrocytes exhibited enhanced tau uptake via heparan sulfate proteoglycan- and lipoprotein receptor-related protein 1-mediated pathways and superior clearance through lysosomal and proteasomal degradation. They exported less tau, limiting propagation. Transcriptomic analyses revealed upregulation of genes involved in cell projection assembly and endocytosis. Lipidomic profiling showed reduced ceramides and gamma-linolenic acid, linked to decreased neuroinflammation and ferroptosis.

DISCUSSION: APOE3-Ch astrocytes promote tau clearance and metabolic adaptations, providing insights into genetic resilience in AD and potential therapeutic targets.

KEYWORDS
Alzheimer's disease, APOE3-Christchurch, endolysosomal trafficking, heparan sulfate proteoglycans, induced pluripotent stem cell–derived astrocytes, lipid metabolism, lipoprotein receptor-related protein 1, lysosomal degradation, proteasomal degradation, tau oligomers, tau pathology

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Highlights

- APOE3-Christchurch (APOE3-Ch) astrocytes exhibit significantly increased tau internalization compared to wild-type astrocytes, facilitated by upregulated heparan sulfate proteoglycan and low-density lipoprotein receptor-related protein 1 pathways.
- APOE3-Ch astrocytes demonstrate more efficient tau degradation via both lysosomal and proteasomal pathways, while exporting significantly less tau, potentially reducing tau propagation in the central nervous system.
- APOE3-Ch astrocytes show upregulation of genes involved in cell projection assembly and endocytosis, suggesting structural and functional modifications that enhance tau processing.
- Lipidomic profiling reveals reduced ceramide levels and gamma-linolenic acid downregulation in APOE3-Ch astrocytes, alterations linked to reduced neuroinflammatory and ferroptotic activity, contributing to the protective phenotype.

1 | BACKGROUND

Astrocytes are critical for maintaining central nervous system (CNS) homeostasis, playing essential roles in lipid metabolism and the clearance of toxic protein aggregates. In addition to regulating extracellular ion and neurotransmitter levels, astrocytes metabolize excess lipids from neurons during periods of heightened neuronal activity, preventing lipotoxicity and supporting neuronal function.[1,2] Central to these processes is apolipoprotein E (apoE), a lipid-binding protein expressed by astrocytes, which facilitates cholesterol transport, extracellular amyloid beta (Aβ) clearance, synaptic pruning, and blood–brain barrier (BBB) integrity.[2] The functional diversity of *APOE* isoforms profoundly impacts astrocyte biology.[3–5]

While extensive research has focused on the common *APOE* isoforms (*APOE* ε2, *APOE* ε3, and *APOE* ε4), much less is known about rare variants such as APOE3-Christchurch (APOE3-Ch). APOE3-Ch has been associated with reduced Alzheimer's disease (AD) risk and neuroprotection in individuals carrying high-risk genetic mutations for AD.[6–10] However, the underlying mechanisms of protection remain poorly understood.

In addition to their role in lipid metabolism, astrocytes are increasingly recognized as active participants in tau clearance, a process critical for limiting the spread of pathological tau aggregates in AD and related tauopathies.[11–13] Pathological tau, characterized by hyperphosphorylation and aggregation into neurofibrillary tangles, disrupts synaptic integrity and neuronal connectivity, driving neurodegeneration and cognitive decline.[14,15] The mechanisms of tau uptake in astrocytes remain underexplored. The efficiency of tau processing is influenced by *APOE* isoforms, with *APOE* ε4 impairing tau uptake and trafficking, potentially exacerbating tau pathology.[16]

To explore the protective mechanisms conferred by APOE3-Ch, this study investigates the lipid metabolism and tau processing dynamics of APOE3-Ch astrocytes compared to wild-type (WT) astrocytes. Using induced pluripotent stem cell (iPSC)–derived astrocytes, we combined lipidomic profiling, transcriptomic analysis, and functional assays to characterize their phenotypic and functional responses. By delineating the unique lipid and tau regulatory mechanisms of APOE3-Ch astrocytes, this study offers novel insights into the molecular underpinnings of astrocyte-mediated neuroprotection. These findings highlight the potential of targeting astrocytic pathways as a therapeutic strategy for AD and related tauopathies.

2 | METHODS

2.1 | Experimental design

This study investigates the protective mechanisms conferred by the APOE3-Ch variant in the context of AD, focusing on astrocytic lipid metabolism and tau processing. To achieve this, human iPSC (hiPSC)–derived astrocytes from a WT line and two lines homozygous for the APOE3-Ch variant were used. The objectives included characterizing differences in lipid metabolism, elucidating the mechanisms underlying tau uptake and clearance, and identifying transcriptomic changes linked to protective pathways.

Lipid metabolism was analyzed using a combination of lipidomic profiling, lipid droplet induction, and apoE secretion assays. Lipid droplet formation was induced using oleic acid (OA), and the resulting droplets were quantified using confocal microscopy and computational image analysis. Lipidomic profiling was performed on cell and media samples to assess intracellular and secreted lipid species, and *apoE* secretion was quantified using enzyme-linked immunosorbent assay (ELISA).

Tau processing and degradation mechanisms were studied using purified, fluorescently labeled tau oligomers to assess uptake and degradation pathways. Competition assays with heparin and

RESEARCH IN CONTEXT

1. **Systematic review**: The authors reviewed the literature using traditional sources (e.g., PubMed) to evaluate the role of astrocytes in tau pathology and Alzheimer's disease (AD). Studies have shown that astrocytes contribute to both tau clearance and propagation, with receptor-mediated pathways such as lipoprotein receptor-related protein 1 and heparan sulfate proteoglycans playing critical roles. However, the effects of genetic variants like APOE3-Christchurch (APOE3-Ch) on astrocytic tau processing remain poorly understood. Recent reports of APOE3-Ch conferring resilience to AD motivated this study, which examines its influence on astrocytic tau handling and lipid metabolism.
2. **Interpretation**: Our findings provide mechanistic insights into how APOE3-Ch astrocytes enhance tau clearance and limit tau propagation through upregulated endocytosis, lysosomal degradation, and lipid metabolic adaptations. These results suggest that astrocytic functions modulated by APOE3-Ch contribute to neuroprotection in AD.
3. **Future directions**: Future studies should assess whether APOE3-Ch astrocytes confer similar protective effects in vivo, investigate the mechanistic link between altered lipid metabolism and tau clearance, and explore therapeutic strategies to mimic APOE3-Ch's effects in sporadic AD cases.

receptor-associated protein (RAP) determined the roles of heparan sulfate proteoglycans (HSPGs) and lipoprotein receptor-related protein 1 (LRP1) in tau uptake. Degradation pathways were further investigated using lysosomal and proteasomal inhibitors, followed by flow cytometric quantification of tau clearance.

Transcriptomic profiling was conducted to identify the molecular pathways associated with APOE3-Ch astrocytes. RNA sequencing was performed on astrocytes under basal conditions and after tau treatment, enabling differential expression analysis and pathway enrichment analysis.

All experiments were conducted in biological triplicate or quadruplicate to ensure robustness and reproducibility.

2.2 | Ethical statement

Our research complies with all relevant ethical regulations set by the University of California Santa Barbara, and study procedures were approved by the institutional review board (#767).

2.3 | hiPSC culture

hiPSC lines, including a WT parental line (KOLF2.1J, Cat. No. JIPSC1000) and two lines homozygous for the APOE3-Ch mutation (denoted as 1262 and 1264, Cat. No. JIPSC1262 and JIPSC1264), were obtained from The Jackson Laboratory. Cells were cultured in Matrigel-coated (Corning, Cat. No. 354277), tissue culture–treated plates using mTeSR Plus medium (Stemcell Technologies, Cat. No. 100-0276). The medium was replaced every other day until the cells reached ≈ 70% confluence. Cells were passaged at a 1:20 ratio using ReLeSR (Stemcell Technologies, Cat. No. 100-0483), following the manufacturer's protocol. Cells were cryopreserved in mTeSR Plus medium with 10% DMSO (Sigma-Aldrich, Cat. No. D8418).

2.4 | Differentiation of iPSCs into astrocytes

iPSC-derived astrocytes were differentiated from both WT and APOE3-Ch iPSCs using a modified version of previously published protocols[17, 18] dissociated using Accutase (Stemcell Technologies, Cat. No. 07922) and seeded into 24-well AggreWell 800 plates (Stemcell Technologies, Cat. No. 34815) at a density of 2 million cells per well in neural induction medium, which consists of DMEM/F-12 (Gibco, Cat. No. 11320033), 1× B-27 supplement minus vitamin A (50X, Gibco, Cat. No. 12587010), 1× N-2 supplement (100X, Gibco, Cat. No. 17502048), and 10 nM Y-27632 dihydrochloride or ROCK inhibitor (Stemcell Technologies, Cat. No. 72304). This setup was used to generate embryoid bodies (EBs). On Day 2, the medium was replaced with the neural induction medium containing 0.1 μM LDN193189 (Tocris, Cat. No. 6053) and 10 μM SB431542 (Tocris, Cat. No. 1614), while the ROCK inhibitor was removed. Half media changes were performed every other day.

By Day 7, EBs exhibited clear and circular borders. The EBs were collected and plated onto Matrigel-coated plates. Adherent cultures were maintained with the same media and fed every other day for an additional week. On Day 14, neural rosettes were isolated using STEMdiff Neural Rosette Selection Reagent (Stemcell Technologies, Cat. No. 05832) and replated on Matrigel-coated plates in neural progenitor cell (NPC) medium, which consists of DMEM/F12, 1× B27 minus vitamin A, 1× N2 supplement, and 20 ng/mL FGF-2 (Thermo Fisher Scientific, Cat. No. PHG0367) in 1% bovine serum albumin (BSA; Sigma-Aldrich, Cat. No. A9418). Full media changes were conducted every other day until confluency was reached.

Around Day 21, NPCs were harvested and stained with PE-conjugated anti-CD133/1 antibody (Miltenyi Biotec, Cat. No. 130-113-670) and PerCP-Cy5.5-conjugated anti-CD271 antibody (BD Biosciences, Cat. No. 560834). $CD133^+/CD271^-$ populations were sorted using a SONY MA900 cell sorter, with mouse IgG1κ (BD Biosciences, Cat. No. 550795) as an isotype control.

$CD133^+/CD271^-$ NPCs were expanded in NPC media and cryopreserved in NPC media supplemented with 10% DMSO. Additionally, cells were plated on Matrigel-coated plates in Astrocyte Media (ScienCell Research Laboratories, Cat. No. 1801) for astrocyte

differentiation. The cells were cultured in Astrocyte Media for 2 weeks before conducting cellular assays. Astrocytes were cryopreserved in Astrocyte Media with 10% DMSO for later use.

2.5 | RNA extraction and sequencing

Differentiated astrocytes were plated in 60 mm dishes and cultured until they reached full confluency. Three biological replicates were prepared for each cell line, including three replicates for WT astrocytes and three replicates each for the 1262 and 1264 cell lines. For RNA extraction, cells were lysed with TRIzol reagent (Invitrogen, Cat. No. 15596026) according to the manufacturer's instructions. Cold chloroform (Sigma-Aldrich, Cat. No. 366927) was added to the TRIzol cell suspension at a ratio of 1:5. The mixture was vortexed for 15 seconds and then incubated at room temperature for 3 minutes to allow for phase separation. After incubation, the sample was centrifuged at 12,000 × g for 15 minutes at 4°C. The transparent aqueous (upper) phase was carefully transferred to a new tube, ensuring no disruption of the interphase layer. An equal volume of isopropanol (Sigma-Aldrich, Cat. No. 34863) was added to the aqueous phase, and the tube was gently inverted five times to mix. The mixture was then incubated at room temperature for 10 minutes, followed by centrifugation at 12,000 × g for 10 minutes at 4°C. The supernatant was discarded. The RNA pellet was washed by adding an equal volume of cold ethanol (Sigma-Aldrich, Cat. No. 459828), followed by brief vortexing. The sample was then centrifuged at 7500 × g for 5 minutes at 4°C. The supernatant was thoroughly removed without disturbing the RNA pellet. The RNA pellet was resuspended in UltraPure DNase/RNase-free distilled water (Invitrogen, Cat. No. 10977015) and incubated at 55°C for 10 minutes with gentle vortexing every 5 minutes to ensure complete dissolution.

Library preparation for RNA sequencing using NEBNext Ultra II Directional RNA Library Prep Kit (New England Biolabs, Cat. No. E7760S) with the NEBNext Poly(A) mRNA Magnetic Isolation Module (New England Biolabs, Cat. No. E7490S) and NEBNext Multiplex Oligos for Illumina (New England Biolabs, Cat. No. E7335S) and next-generation sequencing were performed at the University of California Santa Barbara (UCSB) Biological Nanostructures Laboratory. Sequencing was done on an Illumina NextSeq 500 using a 75-cycle high-output kit.

2.6 | Transcriptomic analysis

RNA sequencing fastq files were quality-checked using FastQC and trimmed with Trimmomatic. The trimmed fastq files were then aligned to the GRCh38 human genome assembly with HISAT2. FeatureCounts was used to generate a count matrix by counting gene features. Differential expression analysis was conducted with the DESeq2 pipeline, which combines the two APOE3-Ch lines to represent a single genotype and compares phenotypic differences between genotypes. Gene counts were normalized using the DESeq2 rlog function for regularized log transformation.

2.7 | Lipid droplet induction and quantification

To induce lipid droplet production and assess lipid buffering capacity in astrocytes, OA (Sigma-Aldrich, Cat. No. O1008) was dissolved in a fatty acid (FA)-free BSA solution in 0.1 M Tris, pH 8.0 (Invitrogen, Cat. No. AM9855G), following a previously published protocol.[19] Differentiated astrocytes were plated at a density of 20,000 cells per well on Matrigel-coated 96-well glass-bottom plates in Astrocyte Media. Cells were incubated with 20 μM OA in BSA at 37°C for 18 hours. After incubation, cells were fixed with 4% paraformaldehyde and stained with LipidTOX Deep Red (Invitrogen, Cat. No. H34477, 1:1000) and Hoechst 33342 (same as above, 1:10,000) at room temperature for 30 minutes. After staining, cells were washed twice with phosphate-buffered saline (PBS) and stored in PBS until imaging. Imaging was conducted with a Leica SP8 resonant scanning confocal microscope at the UCSB NRI-MCDB Microscopy Facility, using a 60X objective.

Lipid droplet quantification was conducted using a puncta identification algorithm. Puncta centers were detected using the blob_log function from the scikit-image library (version 0.24.0), which implements the Laplacian of Gaussian method for blob detection. The minimum and maximum size parameters for puncta were determined based on manual measurements performed in ImageJ. To ensure detection quality, each identified punctum was enclosed within a bounding box of 2σ width (where σ values for each punctum are provided by blob_log), and puncta with a signal-to-noise ratio $< 1\%$ were excluded to remove low-quality detections. The shape of each punctum was further refined using a flood-fill function applied to the inverted intensity values within the bounding box. This approach enabled the extraction of detailed shape characteristics, including area, eccentricity, and peak intensity.

2.8 | *apoE* ELISA

One hundred twenty thousand differentiated astrocytes were plated in Matrigel-coated 24-well plates prior to collection. On the collection day, media were harvested and centrifuged at 1500 x g for 5 minutes to remove cell debris, with the supernatants retained as media samples. Cell pellets were then lysed in radioimmunoprecipitation assay (RIPA) buffer (Thermo Fisher Scientific, Cat. No. 89900) supplemented with ethylenediaminetetraacetic acid (EDTA)-free cOmpleteTM protease inhibitor cocktail (Roche, Cat. No. 11836170001) and 1 mM phenylmethylsulfonyl fluoride (PMSF; 0.1 M, Boston BioProducts, Cat. No. BP-481). The total protein concentration of both cell pellet samples and media samples was determined using the PierceTM BCA protein assay kit (Thermo Fisher Scientific, Cat. No. 23225). *apoE* quantification was performed using the Human APOE ELISA kit (Invitrogen, Cat. No. EHAPOE), following the manufacturer's protocol. The normalized

apoE concentration for both cell pellet and media samples were calculated by normalizing the *apoE* concentration measured by ELISA to the total protein concentration of the corresponding cell pellet sample.

2.9 | Cell viability assay

Differentiated astrocytes were plated in a Matrigel-coated 96-well plate at a density of 20,000 cells per well. Control wells containing medium without cells were included to determine background luminescence. The next day, cells were treated with 20 μM OA in BSA for 18 hours. Cell viability was assessed using the CellTiter-Glo Cell Viability Assay (Promega, Cat. No. G7570) according to the manufacturer's instructions. Briefly, the plate and its contents were equilibrated to room temperature, and CellTiter-Glo Reagent was added directly to wells containing medium. The plate was then mixed on a shaker for 2 minutes to induce cell lysis and incubated at room temperature to stabilize the luminescent signal. Luminescence was measured using a Tecan Spark 10 M Multimode Plate Reader.

2.10 | Lipidomics

Differentiated astrocytes were plated in 10 cm dishes and cultured to full confluency. Four biological replicates were prepared for each cell line, including four replicates for WT astrocytes and four replicates each for the 1262 and 1264 cell lines. Media were collected and centrifuged at 1500 x g for 5 minutes. Cell pellets were dissociated with Accutase by incubation at 37°C for 10 minutes, followed by centrifugation at 2000 x g for 5 minutes at 4°C. Both media samples and cell pellets were frozen at −80°C and lyophilized. The lyophilized samples were processed at the University of Texas Medical Branch Mass Spectrometry Facility for targeted lipidomics of neutral lipids, polar lipids, and FAs.

For sample processing, isotopically labeled lipid standards (UltimateSPLASH ONE) and free FAs (FFAs) were added to each sample prior to extraction. Samples were extracted with hexane (3:2, v/v), centrifuged, and the supernatant was evaporated under nitrogen. Dried samples were resuspended in 100 μL dichloromethane/methanol (1:1, v/v). Protein concentration of the remaining cell pellets was determined using a bicinchoninic acid (BCA) assay. For lipidomics, 30 μL of each sample extract was transferred into a glass autosampler vial for neutral lipid analysis, and 50 μL was dried, resuspended in 50 μL of 80% methanol, and derivatized with 3-NPH.

Liquid chromatography (LC) tandem mass spectrometry analysis was performed using an Acquity H-class HPLC System (Waters) coupled to a QTRAP 6500 mass spectrometer (SCIEX). Neutral lipids were separated by reverse-phase chromatography on an Accucore C30 column (Thermo Scientific). For triacylglycerols (TGs) and determination of their FA compositions, the pooled quality control (QC) was injected multiple times with different survey methods to cover TG FA carbon numbers from 44 to 60. FAs analyzed included 14:0, 15:0, 16:0, 16:1, 17:0, 18:0, 18:1, 18:2, 18:3, 19:0, 20:0, 20:1, 20:2, 20:3, 20:4, 20:5, 22:4, 22:5, 22:6, 24:0, and 24:1. A Python script (v3.11.5) generated all possible TG combinations, providing precursor and product ion masses based on FA fragmentation. Overlapping TG FA peaks were curated into a final, sample-specific method to determine the complete FA composition for each TG molecular species in the lipid extracts.

2.11 | Lipidomic analysis

LC mass spectrometry peak areas were integrated using MultiQuant (v3.0.3, SCIEX). FFA peak areas were then exported and normalized to the peak area its respective [13C6]-3NPH internal standard and protein concentration of each sample (pellet samples only) using a custom R script (R v4.3.3, RStudio v2023.12.1) to give relative values of "Normalized Peak Area / mg protein."

Prior to data analysis, a data QC analysis was performed using a custom R Markdown document to assess the response from internal standards within samples and the pooled QC, injection order effects, retention time variability, and variability of biological lipids.

To prepare the data for statistical testing, samples were organized by genotype and replicate. A design matrix was created to compare the APOE3-Ch and WT astrocyte genotypes, and columns with zero variance were removed to ensure robust statistical analysis. Differential expression analysis was conducted using the limma package. A linear model was fitted to the data, followed by the construction of a contrast matrix to compare the APOE3-Ch samples to WT. Empirical Bayes moderation was applied to the contrast-adjusted model, and results were extracted, adjusting P values using the Holm method. A threshold of $|\log FC| > 0.1$ and $P < 0.05$ was applied to identify statistically significant lipid species.

2.12 | Tau purification

Full-length tau protein with the P301L point mutation (2N4R-P301L) was purified following a previously published protocol.[20] Briefly, 2N4R-P301L tau encoded in the pRK172 plasmid was expressed in *E. coli* BL21 (DE3). Cell pellets were collected and resuspended in a lysis buffer containing 50 mM MES (pH 6.5), 5 mM dithiothreitol (DTT), 1 mM PMSF, and 1 mM egtazic acid (EGTA), supplemented with cOmplete protease inhibitor tablets. The cell lysate was sonicated, boiled for 10 minutes, and centrifuged at 50,000 × g for 30 minutes at 4°C. The supernatant was precipitated with 20% (w/v) ammonium sulfate and centrifuged at 20,000 × g for 30 minutes at 4°C. The resulting pellet was resuspended in 4 mL of Buffer A (50 mM MES, pH 6.5, 50 mM NaCl, 2 mM DTT, 1 mM PMSF, 1 mM EGTA) and dialyzed overnight against the same buffer. The protein was then loaded onto a HiTrap SP HP column (Cytiva, Cat. No. 17115201) and eluted using a linear NaCl gradient with Buffer B (Buffer A + 1 M NaCl). Fractions containing 2N4R-P301L tau were pooled, concentrated, and dialyzed overnight into PBS (pH 7.4). Protein concentration was determined using a BCA assay.

2.13 | Tau labeling

2N4R-P301L tau were labeled with Alexa FluorTM 647 NHS ester (Invitrogen, Cat. No. A20006) according to the manufacturer's instructions. After labeling, 100 mM glycine was added to quench the reaction, and unreacted dye was removed using Zeba desalting columns (Thermo Scientific, Cat. No. 89882). Average label incorporation was ≈ 1.5 moles/mole of protein, as determined by measuring fluorescence and protein concentration (A_{max} x MW of protein/[protein] x ε_{dye}).

2.14 | Tau fibrillization

To prepare tau oligomers, monomeric 2N4R-P301L tau protein was aggregated following a previously published protocol.[20] Briefly, 10 μM tau protein in PBS with 1 mM DTT (pH 7.4) was combined with 0.05 mg/mL heparin and incubated at 37°C with shaking for 4 hours. The resulting tau oligomers were buffer exchanged into PBS using Amicon Ultra Centrifugal Filter (Millipore, Cat. No. UFC500308) to remove excess DTT and heparin.

2.15 | Tau uptake assay

Astrocytes were seeded at a density of 20,000 cells per well in a 24-well plate pre-coated with Matrigel and maintained in Astrocyte Media. After 24 hours, the media were replaced, and the cells were treated with 50 nM of 2N4R-P301L tau oligomers conjugated to Alexa Fluor 647 (AF647) for 3 hours at 37°C. After treatment, the cells were washed twice with PBS to remove any non-internalized tau and then dissociated using Accutase for flow cytometry analysis.

Dissociated cells were analyzed using a SONY MA900 cell sorter, and median AF647 fluorescence intensity was measured to quantify tau uptake. The median fluorescence intensity of each cell line treated with tau was normalized to its respective control condition (PBS treatment).

2.16 | Heparin competition assay

For heparin competition, astrocytes were pre-treated with media containing heparin at 0, 1, 50, and 100 μg/mL for 16 hours prior to tau treatment. After this incubation, cells were washed twice with PBS, and fresh media containing 50 nM tau oligomers was added. The subsequent steps for the tau uptake assay were carried out as described above.

2.17 | RAP purification

RAP was cloned into the pMCSG7 vector and purified using nickel-nitrilotriacetic acid (Ni-NTA) agarose (Qiagen, Cat. No. 30210). The RAP construct was transformed into Escherichia coli DH5α chemically competent cells (Thermo Fisher Scientific, Cat. No. 18265017). Cells were grown to an optical density (OD600) of 0.6, at which point protein expression was induced with isopropyl β-D-1-thiogalactopyranoside (IPTG).

For purification, Ni-NTA agarose was prepared with HIS binding buffer (50 mM Tris-HCl, pH 8.0, 1 M NaCl, 10 mM imidazole). Non-specific proteins were removed using HIS wash buffer (50 mM Tris-HCl, pH 8.0, 300 mM NaCl, 30 mM imidazole). RAP protein was eluted with HIS elution buffer containing 50 mM Tris-HCl, pH 8.0, 300 mM NaCl, and 300 mM imidazole. The protein concentration was quantified using the Pierce BCA protein assay (Thermo Scientific, Cat. No. 23228).

2.18 | RAP competition assay

For RAP competition, astrocytes were treated with recombinant RAP at 0, 50, 100, and 500 nM concurrently with tau addition. The protocol for tau uptake quantification was then performed as outlined in tau uptake assay.

2.19 | Transmission electron microscopy

To prepare samples for transmission electron microscopy (TEM) analysis, 5 μL of tau aggregate solution was applied to a glow-discharged copper grid (Electron Microscopy Sciences, Cat. No. FCF-200-Cu) and allowed to adsorb for 20 seconds. Excess liquid was gently removed by blotting with filter paper. The grids were then stained with 5 μL of 1.5% (w/v) uranyl acetate solution (Electron Microscopy Sciences, Cat. No. 22400-2), followed by immediate blotting to remove excess stain. A second application of 5 μL uranyl acetate was applied for 60 seconds, after which the grid was blotted dry.

The prepared grids were imaged using a Thermo Scientific Talos G2 200X TEM/STEM microscope operating at 200 kV under room temperature conditions. Images were captured using a Ceta II CMOS 4k x 4k camera at specified magnifications to visualize the tau oligomers.

2.20 | Tau clearance assay

Astrocytes were seeded at a density of 120,000 cells per well in a 24-well plate pre-coated with Matrigel and maintained in Astrocyte Media. After 24 hours, the media were replaced, and the cells were treated with 100 nM 2N4R-P301L tau oligomers conjugated to Alexa Fluor 647 (AF647) for 3 hours at 37°C. After treatment, the cells were washed twice with PBS to remove any non-internalized tau. Fresh media were added to the cells, which were then incubated for 0, 6, 12, 24, and 48 hours. At each time point, cells were dissociated using Accutase, fixed with 4% paraformaldehyde at room temperature for 15 minutes, and stored in PBS at 4°C until flow cytometry analysis. Media samples were also collected at each time point, centrifuged at 1500 × g for 5 minutes to remove cellular debris, and the supernatant was stored at 4°C until analysis.

Flow cytometry was performed using a SONY MA900 cell sorter. For cell samples, gating was applied to exclude debris and isolate singlet populations, and the median AF647 fluorescence intensity of the singlet cells was measured to quantify intracellular tau levels at each time point. For media samples, controls, each with three replicates, were included: (1) a positive control containing astrocyte media with 10 nM 2N4R-P301L AF647, (2) a negative control containing only astrocyte media, and (3) astrocyte media collected from untreated astrocytes. The median AF647 fluorescence intensity for media samples was recorded over a 1-minute acquisition period.

2.21 | Tau ELISA

Astrocytes were seeded at a density of 120,000 cells per well in a 24-well plate pre-coated with Matrigel and maintained in Astrocyte Media. After 24 hours, the media were replaced, and the cells were treated with 25 nM 2N4R-P301L tau oligomers conjugated to Alexa Fluor 647 (AF647) for 3 hours at 37°C. After treatment, the cells were washed twice with PBS to remove any non-internalized tau. Fresh media were added to the cells, which were then incubated for 0 and 24 hours. At each time point, cell pellets were lysed in RIPA buffer supplemented with EDTA-free cOmpleteTM protease inhibitor cocktail and 1 mM PMSF. Media were harvested and centrifuged at 1500 x g for 5 minutes to remove cell debris, with the supernatants retained as media samples. The total protein concentration of both cell pellet samples and media samples was determined using the PierceTM BCA protein assay kit. Total tau quantification was performed using the human total tau ELISA kit (Invitrogen, Cat. No. KHB0041), following the manufacturer's protocol. The normalized tau concentration for both cell pellet and media samples was calculated by normalizing the tau concentration measured by ELISA to the total protein concentration of the corresponding cell pellet sample.

2.22 | Tau colocalization study

Astrocytes were seeded at a density of 20,000 cells per well in a 96-well glass bottom plate pre-coated with Matrigel and maintained in Astrocyte Media. After 24 hours, the media were replaced, and the cells were treated with 50 nM 2N4R-P301L tau oligomers conjugated to Alexa Fluor 647 (AF647) for 3 hours at 37°C. After treatment, the cells were washed three times with PBS to remove any non-internalized tau. Fresh media were added to the cells and were then incubated for 8 and 16 hours, respectively. At each time point, cells were fixed for immunocytochemistry probing for tau colocalization with P62, LAMP1, LC3B, and 20S proteasome.

2.23 | Immunocytochemistry

Differentiated astrocytes (> day 14) were plated in Astrocyte Media on Matrigel-coated 96-well glass-bottom plates (Cellvis, Cat. No. P96-0-N). Cells were fixed with 4% paraformaldehyde (16% solution, Electron Microscopy Sciences, Cat. No. 15700) at room temperature for 15 minutes, followed by three washes with ice-cold PBS (Gibco, Cat. No. 10010049). Cells were then permeabilized with 0.1% Triton X-100 (Sigma-Aldrich, Cat. No. X100-5ML) at room temperature for 10 minutes and blocked with 10% normal goat serum (Thermo Fisher Scientific, Cat. No. 50062Z) at room temperature for 1 hour.

For immunostaining of differentiation markers, primary antibodies, including chicken anti-glial fibrillary acidic protein (GFAP; Abcam, Cat. No. ab4674, 1:500), rabbit anti-GFAP (Abcam, Cat. No. ab7260, 1:500), mouse anti-S100 calcium-binding protein B (S100B; Invitrogen, Cat. No. MA1-25005, 1:500), rabbit anti-octamer-binding transcription factor 4 (OCT4; Abcam, Cat. No. ab19857, 1:500), guinea pig anti-microtubule-associated protein 2 (MAP2; SYSY, Cat. No. 188004, 1:500), and mouse anti-sex-determining region Y box 2 (SOX2; Invitrogen, Cat. No.MA1-014, 1:500) were diluted in 5% normal goat serum and incubated with cells overnight at 4°C on a shaker. After incubation, cells were washed with PBST (PBS + 0.1% Tween 20) three times for 5 minutes each. Secondary antibodies, goat anti-mouse immunoglobulin (Ig)G1 Alexa Fluor 488 (Invitrogen, Cat. No. A-21121, 1:1000), goat anti-chicken IgY Alexa Fluor 647 (Invitrogen, Cat. No. A-21449, 1:1000), donkey anti-rabbit IgG (H+L) Alexa Fluor 488 (Invitrogen, Cat. No. A-21206), and goat anti-guinea pig IgG (H+L) Alexa Fluor 555 (Invitrogen, Cat. No. A-21435) were diluted in 5% normal goat serum. Along with Hoechst 33342 staining (Thermo Fisher Scientific, Cat. No. 62249, 1:10,000, diluted in PBS), secondaries were added to the cells and incubated at room temperature in the dark on a shaker. After incubation, cells were washed with PBST three times for 5 minutes each and stored in PBS prior to imaging.

For immunostaining of P62, LAMP1, LC3B, and 20S proteasome, the same procedure was followed. Primary antibodies were diluted in 5% normal goat serum at specific dilution factors. Mouse anti-SQSTM1/P62 (Abcam, Cat. No. ab109012) was used at a 1:500 dilution, rabbit anti-LAMP1 (Cell Signaling, Cat. No. 9091S) at 1:200, rabbit anti-LC3B (Abcam, Cat. No. ab192890) at 1:500, and rabbit anti-20S proteasome core subunits (Enzo, Cat. No. BML-PW8155-0100) at 1:500. Secondary antibody goat anti-mouse IgG Alexa Fluor 568 (Invitrogen, Cat. No. A-11004) was applied at a 1:2000 dilution in 5% normal goat serum, along with Hoechst stain.

For HSPGs and LRP1, the same procedure was followed. To probe for HSPGs, primary antibody mouse anti-10E4 (Amsbio, Cat. No. 370255-S) was applied at a 1:100 dilution in 5% normal goat serum. Secondary antibody goat anti-mouse IgG Alexa Fluor 568 (Invitrogen, Cat. No. A-11004) was applied at a 1:2000 dilution in 5% normal goat serum, along with Hoechst stain. For probing LRP1, primary antibody rabbit anti-LRP1 (Abcam, Cat. No. ab92544) was applied at a 1:1000 dilution in 5% normal goat serum. Secondary antibody goat anti-mouse IgG Alexa Fluor 568 (Invitrogen, Cat. No. A-11004) was also applied at a 1:2000 dilution in 5% normal goat serum, along with Hoechst stain.

2.24 | Colocalization analysis

For the colocalization analysis between internalized tau oligomers and the specified markers, we generated a mask for the non-tau channel by

smoothing the image with a Gaussian kernel and applying the threshold_triangle function from the scikit-image library. The tau channel mask was created using the puncta identification method described in section 2.7.

To quantify colocalization, the overlap coefficient was calculated as the ratio of the intersection area to the union area of the two masks. Additionally, the Mander coefficient was determined by computing the mean intensity of the non-tau channel within the tau channel mask, normalized by the average intensity of the non-tau channel across the image.

2.25 | Tau degradation inhibitor assay

Astrocytes were seeded at a density of 120,000 cells per well in a 24-well plate pre-coated with Matrigel and maintained in Astrocyte Media. After 24 hours, the media were replaced, and the cells were treated with 50 nM 2N4R-P301L tau oligomers conjugated to Alexa Fluor 647 (AF647) for 3 hours at 37°C. After treatment, the cells were washed three times with PBS to remove any non-internalized tau. Fresh media containing 15 μM MG132 or 0.5 μM Bafilomycin A1 were added to the cells, which were then incubated for 16 hours. Cells were dissociated using Accutase, fixed with 4% paraformaldehyde at room temperature for 15 minutes, and stored in PBS at 4°C until flow cytometry analysis.

Flow cytometry was performed using a SONY MA900 cell sorter. For cell samples, gating was applied to exclude debris and isolate singlet populations, and the median AF647 fluorescence intensity of the singlet cells was measured to quantify intracellular tau levels.

2.26 | Tau treatment RNA sequencing

Differentiated astrocytes were plated in 60 mm dishes and cultured until they reached full confluency. Cells were treated with 50 uM of tau oligomers or vehicle (PBS) for 24 hours before extracting RNA. Four biological replicates were prepared for each condition and cell line. For RNA extraction, a protocol described in the RNA extraction section was followed.

Library preparation for RNA sequencing using NEBNext Ultra II Directional RNA Library Prep Kit (New England Biolabs, Cat. No. E7760S) with the NEBNext Poly(A) mRNA Magnetic Isolation Module (New England Biolabs, Cat. No. E7490S) and NEBNext Multiplex Oligos for Illumina (New England Biolabs, Cat. No. E7335S) and Next-Generation Sequencing were performed at the UCSB Biological Nanostructures Laboratory. Sequencing was done on an Illumina NextSeq 500 using a 75-cycle high-output kit.

2.27 | Tau treatment transcriptomic analysis

RNA sequencing fastq files were quality-checked using FastQC and trimmed with Trimmomatic. The trimmed fastq files were then aligned to the GRCh38 human genome assembly with HISAT2. FeatureCounts was used to generate a count matrix by counting gene features. Low-expression genes were filtered by retaining rows with a minimum of 10 counts to ensure robust analysis.

Differential expression analysis was conducted with the DESeq2. The experimental design formula, "Genotype + Treatment + Genotype:Treatment," was used to assess both independent and interactive effects of genotype and treatment. Genotype (WT or APOE3-Ch) and treatment (PBS or Tau-treated) were set as factors, and reference levels were specified to facilitate comparisons. The DESeq2 pipeline was executed, including normalization and dispersion estimation, followed by differential expression testing. Adjusted *P* values (false discovery rate [FDR]) were used to identify significantly differentially expressed genes (DEGs), with a threshold of FDR ≤ 0.05.

Significant DEGs were further analyzed using gene-set enrichment analysis (GSEA) to identify enriched biological pathways. Log2 fold change values of significant genes were ranked and mapped to the DEGs. Gene Ontology gene sets from the Molecular Signatures Database (MSigDB) were imported, and enrichment was performed using the fgsea package. Pathways with normalized enrichment scores (NES) and FDR ≤ 0.05 were identified as significantly enriched. Upregulated and downregulated pathways were visualized using horizontal bar plots, with pathways ranked by NES. Transparency was scaled to reflect FDR values.

Pathways of interest were selected based on GSEA results. Leading-edge genes contributing to pathway enrichment were extracted, and their expression profiles were visualized. Gene expression data were normalized (*z* scored) across samples to highlight relative differences. Genes were grouped into treatment categories (WT untreated, WT treated, APOE3-Ch untreated, APOE3-Ch treated) to capture regulatory patterns.

2.28 | Statistical analysis

All statistical analyses were conducted using R (v4.3.3), Python (v3.11.5), or GraphPad Prism 10. Details of the specific statistical methods are provided in the corresponding method sections. Data are presented as mean ± standard deviation (SD) unless otherwise stated, with statistical significance defined as $P < 0.05$. All analysis parameters and scripts are available upon request to ensure transparency and reproducibility.

3 | RESULTS

3.1 | Characterization of iPSC-derived APOE3-Ch and WT astrocytes

To examine the phenotypic differences caused by the *APOE3-Ch* variant in astrocytes, We differentiated hiPSCs homozygous for either *APOE3* or *APOE3-Ch* into astrocytes using a dual-SMAD inhibition-directed differentiation protocol.[18] The parental iPSC line carrying

the homozygous *APOE3* is referred to as "P" or WT in this study (Figure 1A). This line was subsequently modified using CRISPR to introduce the homozygous APOE3-Ch mutation. For this study, we selected two CRISPR-edited *APOE3-Ch* clones, labeled "1262" and "1264" (Figure 1A).

After differentiation, we confirmed the expression of mature astrocyte markers S100B and GFAP through immunocytochemistry in both WT and APOE3-Ch astrocytes (Figure 1B). Notably, GFAP staining in iPSC-derived astrocytes showed unexpected nuclear localization. To confirm this observation, we repeated the staining using a second GFAP antibody (ab7260), which yielded a similar pattern (Figure S1 in supporting information). While GFAP is typically localized to the cytoplasm, this nuclear signal may reflect altered intermediate filament distribution during early differentiation. Further investigation is needed to verify this phenomenon and clarify the underlying significance.

To further validate the identity of the differentiated astrocytes, we performed immunocytochemistry for lineage-specific markers (Figure S1). iPSC-derived astrocytes of both WT and APOE3-Ch lines showed strong cytoplasmic staining for the astrocyte marker S100B (Figures 1B and S1). In contrast, pluripotency markers OCT4 and SOX2, which were strongly expressed in undifferentiated iPSCs, were markedly reduced or faint in astrocyte cultures, indicating successful loss of pluripotency. The neuronal marker MAP2 was absent in iPSCs and showed only faint signal in astrocyte cultures, suggesting minimal neuronal contamination. These staining results confirm that the astrocyte cultures are largely astrocytic and have undergone appropriate lineage restriction.

RNA sequencing further validated that these iPSC-derived astrocytes express high levels of key astrocyte markers, such as fibroblast growth factor receptor 3, GFAP, S100B, and solute carrier family 1 member 2 (Figure 1C). Notably, however, these astrocytes exhibited low expression of aquaporin 4 (AQP4), a marker of astrocyte maturation critical for water transport and potassium ion buffering. Reduced AQP4 expression may alter water and ion homeostasis in these cells.

Compared to WT astrocytes, APOE3-Ch astrocytes displayed significantly lower expression of the transcription factor nuclear factor 1 X-type (NFIX; Figure 1C). NFIX plays a critical role in astrocyte differentiation and maturation. Its reduced levels in APOE3-Ch astrocytes suggest that these cells may be less mature than the control line, indicating that the APOE3-Ch variant may impede normal astrocyte maturation. Further investigation into how this variant affects astrocyte development and function is warranted.

3.2 | APOE3-Ch astrocytes exhibit lipid dyshomeostasis

apoE is a key component of lipoprotein particles such as chylomicrons, very-low-density lipoproteins (VLDLs), and high-density lipoproteins (HDLs). In the CNS, apoE facilitates the transport of cholesterol and FAs between cells, maintaining lipid homeostasis. This neuron–astrocyte metabolic coupling, mediated by apoE, plays an essential role in preventing activity-induced lipotoxicity in neurons.[1] Clinical studies have suggested that the Christchurch variant may predispose individuals to hyperlipoproteinemia type III (HLPP3), characterized by elevated blood cholesterol and triglyceride levels, early-onset atherosclerosis, and heart disease.[21]

To explore whether APOE3-Ch astrocytes exhibit lipid metabolism dysfunction, we quantified lipid droplet formation in both untreated astrocytes and those treated with OA. Lipid droplets form to buffer excess intracellular lipids. In the absence of OA treatment, APOE3-Ch astrocytes exhibited a trend toward increased lipid droplet production, though not significantly (Figure 1D,E). However, upon OA treatment, APOE3-Ch astrocytes produced significantly more lipid droplets than WT astrocytes, suggesting a reduced ability to secrete excess lipids. To assess whether this dysregulation is related to apoE secretion, we measured both intracellular and extracellular apoE levels using an ELISA. While intracellular apoE levels did not differ significantly between APOE3-Ch and WT astrocytes (Figure 1F), extracellular apoE levels were significantly lower in APOE3-Ch astrocytes (Figure 1G). This finding indicates a diminished capacity for apoE secretion in APOE3-Ch astrocytes, potentially leading to the intracellular accumulation of lipid droplets. This lipid dyshomeostasis may induce lipotoxicity and disrupt metabolic coupling between astrocytes and neurons. To further investigate whether this impaired lipid buffering induces cellular toxicity, we assessed astrocyte viability with an ATP-based cell viability assay after OA treatment. APOE3-Ch astrocytes displayed decreased viability in the presence of excess lipids compared to WT astrocytes (Figure 1H). These results shed light on how the APOE3-Ch variant may contribute to lipid-related pathology, though additional studies are needed to elucidate the mechanisms underlying reduced apoE export and its effects on lipid metabolism.

3.3 | Lipidomic analysis reveals reduced ceramide levels in APOE3-Ch astrocytes

Astrocytes are the main lipid producers in the brain. To explore the impact of the APOE3-Ch variant on astrocytic lipid metabolism, we performed targeted lipidomic profiling on both the cell pellet fractions and media fractions of astrocytes, aiming to uncover differences in the content of intracellular versus secreted lipids.

The intracellular neutral lipid panel revealed that cholesterol esters (CEs), ceramides (Cer), diacylglycerols (DAGs), and TGs were the more abundant lipid classes overall, while dihydroceramides (DCERs) and monoacylglycerols (MGs) exhibited lower abundances (Figure 2A). Notably, no significant differences were observed in the total abundance of these lipid species between APOE3-Ch astrocytes and WT astrocytes.

Given previous findings of elevated unsaturated TAGs in *APOE* ε4 astrocytes, we examined whether APOE3-Ch astrocytes displayed a similar trend. However, lipidomics analysis of the intracellular neutral lipid profile revealed that APOE3-Ch astrocytes exhibited levels of unsaturated TGs comparable to those observed in WT astrocytes (Figure 2B).

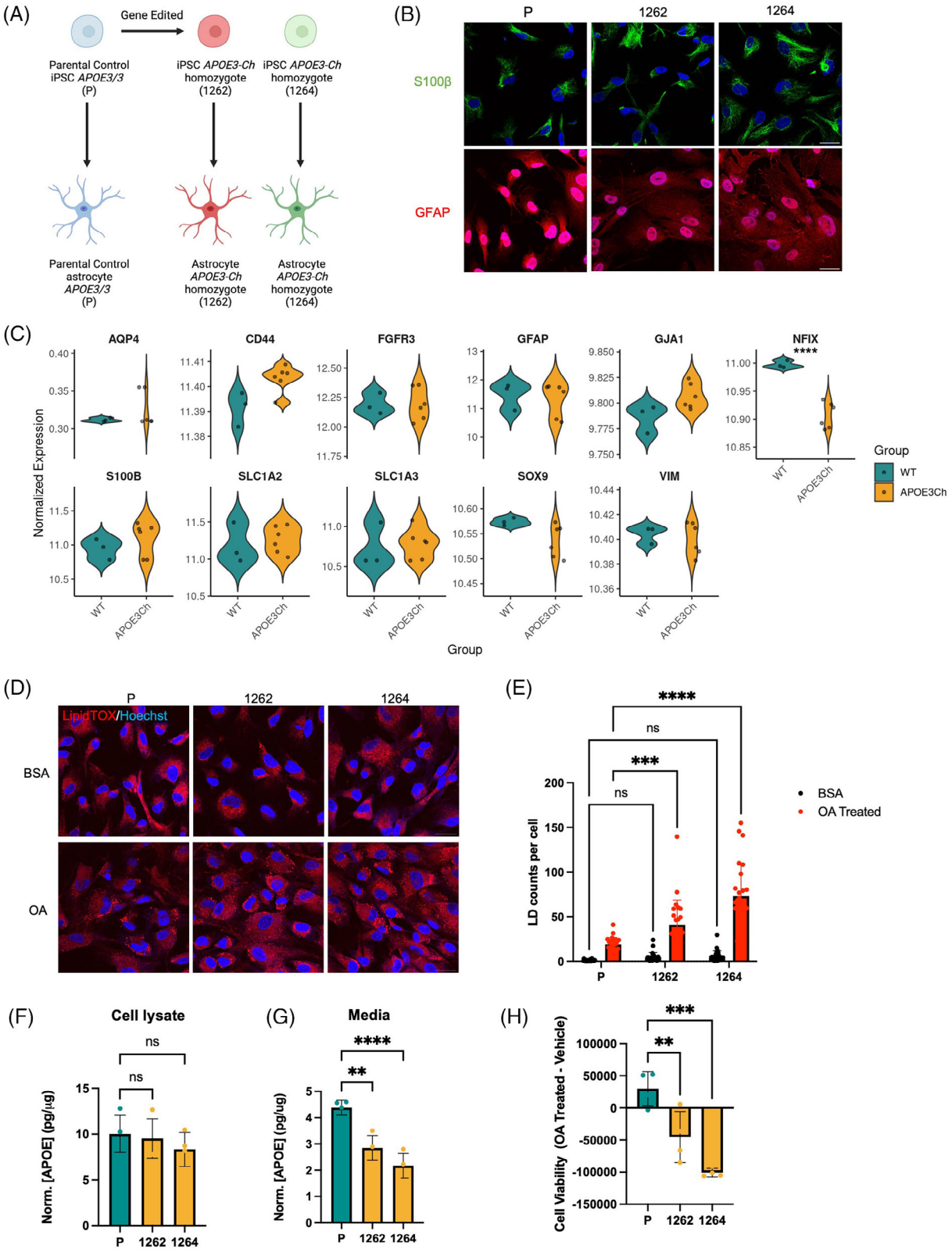


FIGURE 1 Characterization of iPSC-derived astrocytes and their lipid homeostasis. A, Schematic representation of the experimental model. Gene-edited iPSC-derived astrocytes were generated from parental control iPSCs expressing *APOE* ε3/ε3 (P) and from iPSCs homozygous for APOE3-Ch (1262 and 1264). B, Immunofluorescence staining for the astrocytic markers S100B (green) and GFAP (red) in parental (P), 1262, and 1264 astrocytes. Nuclei are counterstained with Hoechst (blue). Scale bar represents 20 µm. C, Violin plots representing normalized expression levels of astrocyte marker genes in APOE3-Ch ($n = 6$) and WT astrocytes ($n = 3$). *P* values were calculated using a Wald test for differential expression between groups (WT vs. APOE3-Ch). Multiple testing correction was performed using the Benjamini–Hochberg method to control the false discovery rate. Genes with an adjusted *P* value < 0.05 were considered statistically significant. D, Lipid droplet (LD) staining using LipidTOX (red) and nuclei staining with Hoechst (blue) in P, 1262, and 1264 astrocytes after treatment with BSA (control) or OA. Scale bar represents 20 µm.

We further analyzed differentially abundant lipid species in both cell pellets and media (Figure 2C). In cell pellets, several Cers and DCERs were significantly reduced, including Cer (14:0), Cer (16:0), Cer (18:0), Cer (18:1), Cer (20:0), Cer (20:1), DCER (14:0), DCER (16:0), and DCER (18:0). Similarly, in the media, Cer (16:0), Cer (18:0), and Cer (22:0) were also significantly reduced. Cers are bioactive sphingolipids involved in cell signaling, apoptosis, and membrane structure, playing a critical role in neurodegenerative diseases like AD.[22, 23] Elevated Cer levels have been associated with AD pathology, including promoting Aβ accumulation, mitochondrial dysfunction, and inflammation.[24, 25] Specifically, increased levels of Cer16, Cer18, Cer20, and Cer24 are found in the brains from patients with AD.[26] Therefore, lower levels of aforementioned Cers, as observed in APOE3-Ch astrocytes, may mitigate these pathological processes, offering a protective effect against AD progression.

To investigate the underlying mechanisms for Cer reduction, we assessed the expression of genes involved in Cer catabolism from the RNAseq data (Figure 2D,E). Several key enzymes, including ACER1, ASAH1, SGMS1, SGMS2, and SPHK1, were significantly upregulated in APOE3-Ch astrocytes. Specifically, SGMS1 and SGMS2, which convert Cers to sphingomyelin, were upregulated. ASAH1 and ACER1 are responsible for converting Cers to sphingosine, while SPHK1 phosphorylates sphingosine to form sphingosine-1-phosphate (S1P), potentially further reducing ceramide levels by diverting sphingosine into the S1P pathway.

We also examined the polar lipid profile (Figure S2 in supporting information) and FA composition (Figure S3 in supporting information) in APOE3-Ch astrocytes. In the polar lipid panel, intracellularly, the most abundant lipids in both WT and APOE3-Ch astrocytes were phosphatidylcholine (PC), phosphatidylethanolamine (PE), phosphatidylinositol (PI), and phosphatidylserine (PS), with no significant differences in total abundance between genotypes (Figure S2A). However, several PE species were notably decreased, while several PS species were increased in APOE3-Ch astrocytes (Figure 2B,C). In the media, differentially abundant polar lipids were scarce, though we observed significant downregulation of PE (14:0_16:0), PE (14:0_20:1), PC (14:0_14:0), and lysophosphatidylcholine (18:2) (Figure S2D,E).

The FA panel also revealed minimal differences in lipid species between APOE3-Ch and WT astrocytes. However, intracellularly, gamma-linolenic acid (GLA) and lauric acid were significantly downregulated (Figure S3A,C) in APOE3-Ch astrocytes. Extracellularly, caproic acid was more abundantly exported in APOE3-Ch astrocytes compared to WT (Figure S3B,C). Notably, dihomo-gamma-linolenic acid (DGLA), a metabolite of the polyunsaturated FA GLA, has been implicated in inducing ferroptosis-mediated neurodegeneration, particularly in dopaminergic neurons.[27] While the role of GLA in astrocytes remains unclear, we hypothesize that the downregulation of GLA in APOE3-Ch astrocytes may help prevent ferroptosis-mediated pathology, contributing to the protective mechanisms associated with this genotype.

Together, these findings suggest a genotype-specific remodeling of lipid metabolism that may contribute to the distinct functional properties of APOE3-Ch astrocytes.

3.4 | APOE3-Ch astrocytes internalize more tau oligomers

In addition to facilitating lipid metabolism, astrocytes also play a critical role in maintaining synaptic homeostasis by clearing harmful materials, including protein oligomers such as tau. Previous studies have demonstrated that astrocytes can internalize both tau monomers and higher-molecular-weight tau species.[13, 28] To investigate how the *APOE3-Ch* variant affects tau uptake, we prepared soluble 2N4R-P301L tau oligomers (Figure S4A in supporting information), labeled them with Alexa Fluor 647 (AF647), and assessed tau uptake in APOE3-Ch and WT astrocytes via flow cytometry (Figure 3A). Our results revealed a significantly increased median AF647 intensity in APOE3-Ch astrocytes, indicating enhanced tau uptake compared to WT astrocytes (Figure 3B,C). This difference became more pronounced with increasing tau concentrations and longer incubation times (Figure S4B,C).

To explore the mechanisms underlying tau uptake in astrocytes, we conducted competition assays using heparin, which inhibits tau uptake by binding to HSPGs (Figure 3D). In WT astrocytes, tau uptake showed only a slight decrease in the presence of heparin, suggesting that HSPGs play a minimal role in mediating tau internalization. The two APOE3-Ch clones (1262 and 1264) exhibited variability in their response to heparin inhibition, possibly due to clonal differences in iPSC lines affecting sensitivity to HSPGs. However, both APOE3-Ch clones inhibited HSPG-mediated uptake when 100 µg/mL of heparin was co-incubated with tau oligomers, suggesting a prominent role of HSPGs in facilitating tau oligomer uptake in APOE3-Ch astrocytes.

We also assessed the role of LRP1 in tau uptake using RAP, an LRP1 antagonist (Figure 3E). RAP treatment drastically reduced tau uptake in both WT and APOE3-Ch astrocytes, illustrating the

E, Quantification of lipid droplet counts per cell in P, 1262, and 1264 astrocytes under BSA or OA treatment conditions ($n = 25$ FOVs taken from 5 biological replicates, 5 FOVs per biological replicate). Significance was determined by two-way ANOVA with Dunnett multiple comparison tests. F,G, *APOE* protein levels normalized to total protein (pg/µg) in cell lysates (F) and media (G) of P, 1262, and 1264 astrocytes ($n = 3$). Significance was determined by one-way ANOVA with Dunnett multiple comparison tests. H, Cell viability after OA)treatment relative to vehicle control in P, 1262, and 1264 astrocytes ($n = 3$). Representative images in (D) were selected to show differences in lipid droplet quantity and may not reflect overall cell number. Cell viability in (H) was assessed using ATP-based luminescent assay, which reflects metabolic activity and may not directly correlate with cell number. All data are expressed as mean ± standard deviation with individual data points shown. The results designated as "ns" are not significant; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$. ANOVA, analysis of variance; *APOE*, apolipoprotein E; APOE3-Ch, APOE3 Christchurch; BSA, bovine serum albumin; FOV, field of view; GFAP, glial fibrillary acidic protein; iPSC, induced pluripotent stem cell; OA, oleic acid; S100B, S100 calcium-binding protein B; WT, wild type

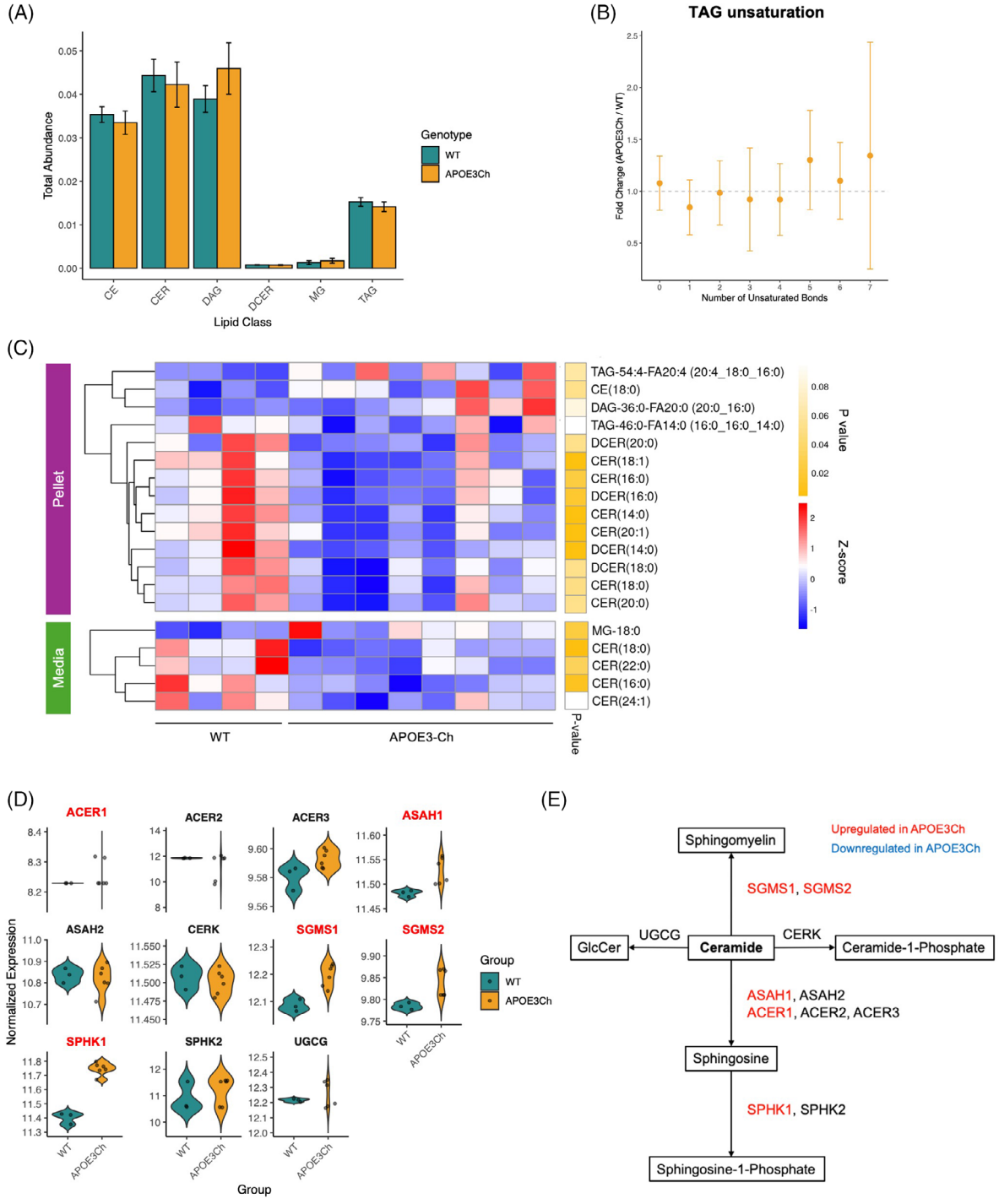


FIGURE 2 The neutral lipid profile suggests reduced ceramide levels in APOE3-Ch astrocytes. A, Total lipid abundance for each lipid species detected in the neutral lipid panel, comparing APOE3-Ch ($n = 8$) and WT astrocytes ($n = 4$). Lipid abundance is represented as the sum of normalized peak area for each lipid class. Error bars denote the SEM. B, Fold change in triacylglycerol (TAG) unsaturation levels between APOE3-Ch and WT astrocytes. The error bars represent the SEM. C, Heatmap showing differentially abundant lipids in the pellets (intracellular) and media (extracellular) of APOE3-Ch astrocytes compared to WT astrocytes. Lipid expression is represented as a *z* score, calculated by subtracting the mean lipid quantity across all samples from the observed quantity and dividing by the standard deviation. Increased and decreased lipids are color-coded. *P* values were calculated using a linear model with empirical Bayes moderation of standard errors. D, Violin plots representing normalized expression levels of genes involved in ceramide catabolism in APOE3-Ch ($n = 6$) and WT astrocytes (n = 3). *P* values were calculated using a Wald test for differential expression between groups (WT vs. APOE3-Ch). Multiple testing correction was performed using the Benjamini–Hochberg method to control the false discovery rate. Genes with an adjusted *P* value < 0.05 were considered statistically significant. Significantly upregulated genes were highlighted in red. E, Schematic representation of the ceramide catabolism pathway. Genes significantly upregulated in APOE3-Ch astrocytes are highlighted in red, while downregulated genes are shown in blue. Downregulated genes were not found. APOE3-Ch, APOE3 Christchurch; SEM, standard error of the mean; WT, wild type

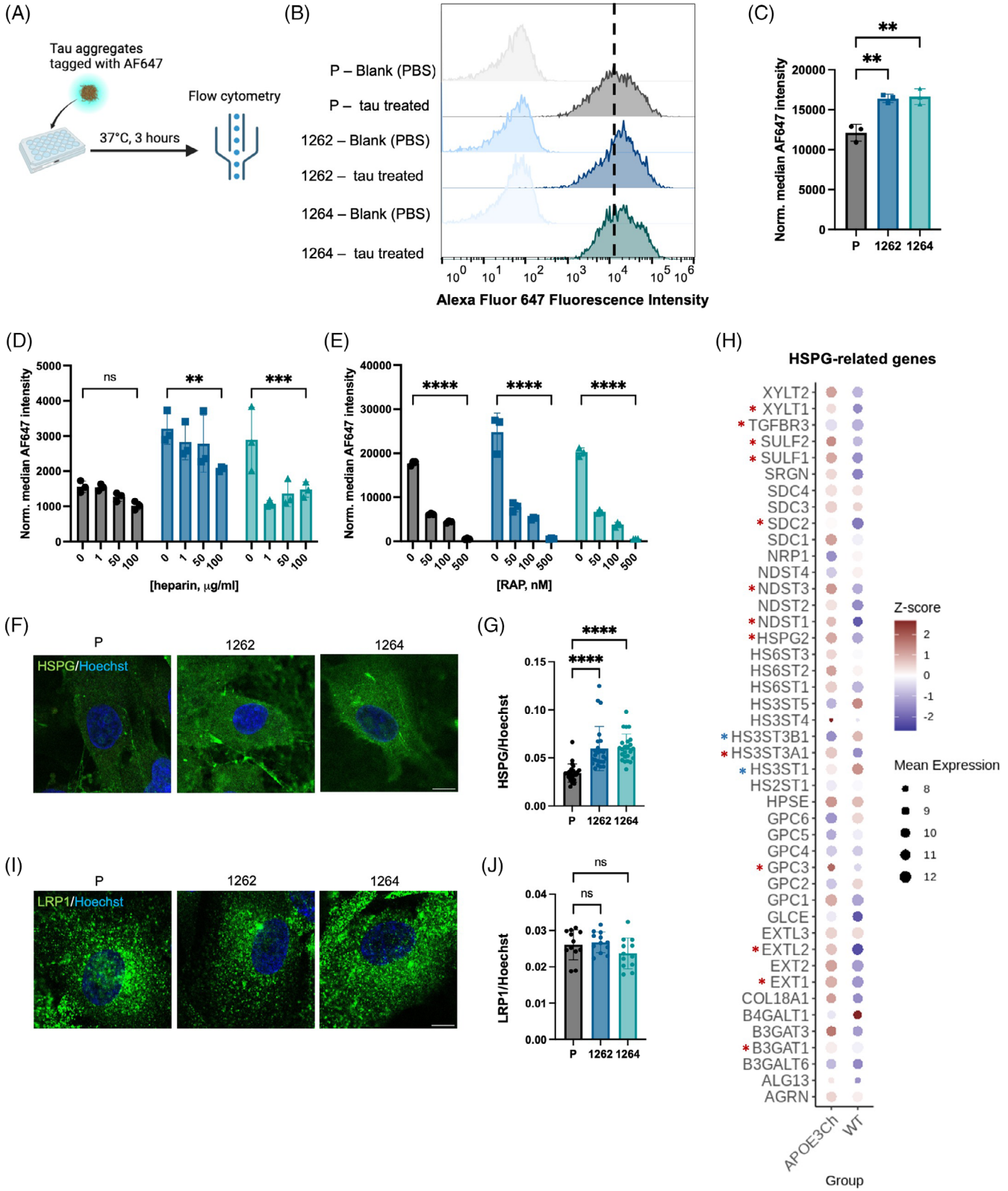


FIGURE 3 APOE3-Ch astrocytes take up more tau aggregates. A, Schematic of the tau uptake assay in iPSC-derived astrocytes. B, Representative overlaid histograms demonstrating the shift in fluorescence of cells with AF647 labeled tau (2N4R-P301L) aggregates (50 nM, 3 h) compared to cells treated with PBS. The dotted line represents the median AF647 intensity in the WT astrocytes treated with tau. C, Measurement of tau uptake by flow cytometry normalized to cells treated with PBS ($n = 3$, 10,000 cells each). D, Measurement of tau uptake in cells treated with tau (50 nM, 3 h) and various concentrations of heparin normalized to cells treated with PBS ($n = 3$, 10,000 cells each). Statistical significance is determined by two-way ANOVA with uncorrected Fisher LSD multiple comparison tests. Displayed is the multiple comparison of cells treated with 100 μg/mL heparin against cells with no treatment for each cell line. E, Measurement of tau uptake in cells treated with tau (50 nM, 3 h) and various concentrations of RAP normalized to cells treated with PBS ($n = 3$, 10,000 cells each). Statistical significance is determined

essential role of LRP1 in tau internalization. While LRP1 is critical in both astrocyte types, HSPG-mediated tau uptake appears more involved in APOE3-Ch astrocytes, potentially contributing to their increased tau internalization.

To further elucidate the mechanisms underlying differential tau uptake, we measured the expression of LRP1 and HSPGs via immunocytochemistry. APOE3-Ch astrocytes exhibited significantly higher HSPG expression, correlating with the greater reliance on HSPG-mediated tau uptake (Figure 3F,G). Corroborating the immunocytochemistry findings, our RNA sequencing data also suggest APOE3-Ch astrocytes increase HSPG expression by significantly upregulating the expression of several HSPG-related genes including *B3GAT1*, *EXT1*, *EXTL2*, *GPC3*, *HS3ST3A1*, *HSPG2*, *NDST1*, *NDST3*, *SDC2*, *SULF1*, *SULF2*, *TGFBR3*, and *XYLT1* (Figure 3H). In contrast, LRP1 expression did not differ significantly between APOE3-Ch and WT astrocytes (Figure 3I,J). These findings suggest that APOE3-Ch astrocytes may facilitate tau uptake through both LRP1- and HSPG-mediated pathways. The increased expression of HSPGs, combined with the use of these dual uptake mechanisms, results in enhanced tau internalization in APOE3-Ch astrocytes compared to WT astrocytes. This differential uptake reflects an intrinsic characteristic of APOE3-Ch astrocytes, potentially contributing to their protective role against AD.

3.5 | APOE3-Ch astrocytes promote tau clearance by degradation, not secretion

After characterizing tau uptake, we investigated tau clearance mechanisms in APOE3-Ch astrocytes. Cells typically clear proteins via degradation or secretion pathways. Based on patient studies showing limited tau pathology in the individual carrying the APOE3-Ch variant and the familial AD mutation PSEN1-E280A,[6] we hypothesized that the APOE3-Ch variant enhances astrocytic tau clearance and thereby limits tau spread in the brain. We conducted tau clearance assays by incubating astrocytes with fluorescently tagged 2N4R-P301L tau oligomers, followed by media change and wash steps to remove non-internalized tau. We then measured intracellular and extracellular tau levels over time via flow cytometry (Figure 4A).

APOE3-Ch astrocytes demonstrated a more pronounced reduction in intracellular tau levels, persisting from 6 to 48 h after media change, suggesting enhanced tau clearance (Figures 4B and S5A in supporting information). To minimize genotype-dependent differences in uptake and focus on degradation kinetics, a higher concentration of tau oligomers (100 nM) was used in this assay. We observed some variability in WT astrocytes at the 0 hour time point, which may reflect technical variation and is noted as a limitation. During the first 12 hours post media change, APOE3-Ch astrocytes exhibited a faster clearance rate compared to WT astrocytes, while the opposite was observed from 12 to 48 hours (Figure S5B,C). This indicates that APOE3-Ch astrocytes activate tau clearance mechanisms more rapidly.

To elucidate the mechanisms behind this rapid clearance, we performed colocalization studies with markers of lysosomal degradation (LAMP1, LC3B, and P62) and proteasomal degradation (20S proteasome). At 8 hours post-media change, APOE3-Ch astrocytes showed increased colocalization of tau with LAMP1, LC3B, and P62, suggesting an early engagement of lysosomal degradation (Figures S6A-D and S7A-H in supporting information). Notably, APOE3-Ch astrocytes showed increased colocalization of exogenous tau with LAMP1, despite having fewer LAMP1+ vesicles than WT astrocytes at 8 hours post-media change (Figure S6A, C, D). This may suggest increased lysosomal turnover or more efficient engagement of tau with lysosomes during the early phase of degradation, though further validation is needed. By 16 hours, the trend of more tau colocalization with lysosomal pathway markers was diminished or reversed. Colocalization with the 20S proteasome followed a different pattern; no significant differences were observed at 8 hours, but increased colocalization was detected in APOE3-Ch (1262) astrocytes at 16 hours (Figure S6E-H). These findings indicate that APOE3-Ch astrocytes may initially rely on lysosomal degradation and efficient lysosomal turnover for tau clearance.

We also examined extracellular tau levels to determine whether enhanced clearance was due to increased secretion (Figure 4C-E). APOE3-Ch astrocytes exported less tau compared to WT astrocytes (Figure 4E), indicating that the enhanced clearance is primarily mediated by degradation rather than secretion.

by two-way ANOVA with uncorrected Fisher LSD multiple comparison tests. Displayed is the multiple comparison of cells treated with 500 nM RAP against cells with no treatment for each cell line. F, Representative images of iPSC-derived astrocytes immunolabeled with anti-10E4, an HSPG marker. Green, 10E4; blue, Hoechst 33342; scale bar = 10 µm. G, Quantification of the expression of HSPGs using fluorescence intensity of anti-10E4 normalized to that of Hoechst per field of view ($n = 24$ FOVs taken from 8 biological replicates, 3 FOVs per biological replicate). H, Quantification of the expression of HSPG-related genes with RNA sequencing ($n = 3$ for WT and $n = 6$ for APOE3-Ch). *Z* score of each HSPG gene is color coded and the size of the circle represents the mean expression. *P* values were calculated using a Wald test for differential expression between groups (WT vs. APOE3-Ch). Multiple testing correction was performed using the Benjamini–Hochberg method to control the false discovery rate. Genes with an adjusted *P* value < 0.05 were considered statistically significant. Significantly upregulated genes were marked by red asterisks and significantly downregulated genes were marked by blue asterisks. I, Representative images of iPSC-derived astrocytes immunolabeled with anti-LRP1. Green, LRP1; blue, Hoechst 33342; scale bar = 10 µm. J, Quantification of the expression of LRP1 using fluorescence intensity of anti-LRP1 normalized to that of Hoechst per field of view ($n = 12$ FOVs taken from 4 biological replicates, 3 FOVs per biological replicate). All data are expressed as mean ± standard deviation with individual data points shown. One-way ANOVA with Dunnett multiple comparison tests was performed to determine the significance unless otherwise specified. The results designated as "ns" are not significant; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$. ANOVA, analysis of variance; APOE3-Ch, APOE3 Christchurch; FOV, field of view; HSPG, heparan sulfate proteoglycan; iPSC, induced pluripotent stem cell; LRP1, lipoprotein receptor-related protein 1; LSD, least significant difference; PBS, phosphate-buffered saline; RAP, receptor-associated protein; WT, wild type

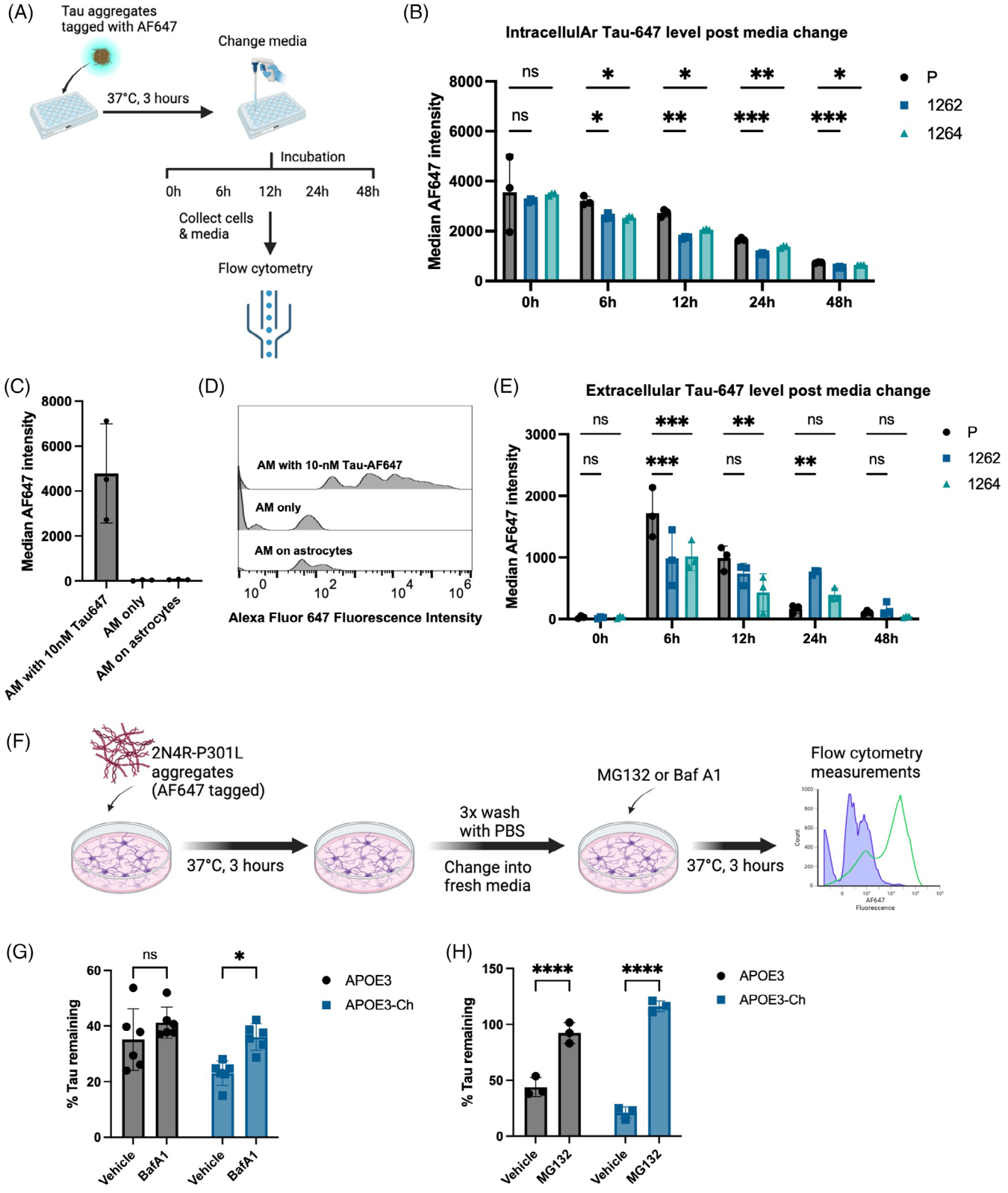


FIGURE 4 APOE3-Ch astrocytes promote more tau degradation and less secretion compared to WT astrocytes. A, Schematic of the tau clearance assay in iPSC-derived astrocytes. B, Measurements of the intracellular tau-AF647 level post media change by flow cytometry at indicated time points (n = 3, 10,000 cells each). Statistical significance is determined by two-way RM ANOVA with Dunnett multiple comparison tests. C, Measurement of the median AF647 intensity in the control samples by flow cytometry. The controls include a positive control, astrocyte media (AM) containing 10 nM of tau-AF647, and two negative controls, astrocyte media only and astrocyte media placed on astrocytes for 1 day ($n = 3$). D, Representative overlaid histograms demonstrating the shift in fluorescence of media containing 10 nM of tau-AF647 compared to the negative controls. E, Measurement of the extracellular tau-AF647 level in the media of cells treated with tau-AF647 (100 nM, 3 hours) after media change at 0 , 6, 12, 24, and 48 hours ($n = 3$). Statistical significance is determined by two-way RM ANOVA with Dunnett multiple comparison tests.

We further validated these findings using tau degradation inhibition assays with MG132, a proteasome inhibitor, and Bafilomycin A1 (BafA1), a lysosome inhibitor (Figure 4F-H). After 16 hours of incubation with MG132, both WT and APOE3-Ch astrocytes showed a halt in tau clearance, indicating that proteasomal degradation is essential for tau clearance in astrocytes (Figure 4H). Interestingly, BafA1 treatment had minimal impact on tau clearance in WT astrocytes, while it partially inhibited tau clearance in APOE3-Ch astrocytes, suggesting that lysosomal degradation contributes more specifically to the enhanced tau clearance observed in APOE3-Ch astrocytes (Figure 4G). We acknowledge that the effect size observed when comparing BafA1-treated and untreated APOE3-Ch astrocytes is small. This may reflect the relatively low concentration of BafA1 used or suggest that tau oligomers in these cells are primarily processed through proteasomal rather than lysosomal pathways. Nonetheless, the modest reduction in tau clearance after BafA1 treatment may indicate a limited but detectable role for lysosomal degradation in APOE3-Ch astrocytes, though further studies are needed to clarify this contribution.

Finally, we confirmed these results using total tau ELISA at representative time points (Figure S8A,B in supporting information). After 24 hours of clearance, the total tau concentration in APOE3-Ch astrocytes decreased ≈ 2-fold, whereas it did not change significantly in WT astrocytes. Additionally, the extracellular tau concentration in APOE3-Ch astrocytes was lower than that in WT astrocytes after 24 hours, supporting our observation that APOE3-Ch astrocytes favor degradation over secretion and tau is less likely to spread from APOE3-Ch astrocytes.

Overall, while proteasomal degradation is critical in both APOE3-Ch and WT astrocytes, there is no evidence to suggest that proteasomal activity differs significantly between the two astrocyte types. These results suggest that the enhanced tau clearance observed in APOE3-Ch astrocytes is partially due to their effective use of lysosomal pathways.

3.6 | Transcriptomic analysis unveils enhanced cell projection and endocytosis in APOE3-Ch astrocytes for tau clearance

To investigate the effects of excess tau on astrocytes and the differences between WT and APOE3-Ch astrocytes at both basal levels and upon tau treatment, we performed RNA sequencing on four groups: WT untreated, WT treated with tau, APOE3-Ch untreated, and APOE3-Ch treated with tau (Figure 5A).

Differential gene expression analysis provided insights into how tau oligomers impacted WT and APOE3-Ch astrocytes, revealing distinct responses between the two genotypes. First, to understand the effects of tau treatment on WT astrocytes, we compared the DEGs between WT-treated and WT-untreated astrocytes (Figure 5B). GSEA indicated that the top-upregulated pathways were primarily involved in the cell cycle process (Figure 5C), potentially indicating a stress response leading to increased cellular proliferation as repair mechanisms in WT astrocytes under tau stress.

Next, we analyzed the transcriptomic response of APOE3-Ch astrocytes to tau treatment by comparing DEGs in tau-treated versus untreated APOE3-Ch astrocytes (Figure S9A,B in supporting information). Enrichment analysis suggested that APOE3-Ch astrocytes under tau stress downregulate pathways related to the negative regulation of responses to stimuli, potentially indicating increased sensitivity and responsiveness to external cues.

Given our previous findings that APOE3-Ch astrocytes process tau differently from WT astrocytes, we investigated the genotype–treatment interaction in the RNA sequencing data to compare their transcriptomic responses to tau treatment (Figure 5D,E). The genotype–treatment interaction term measures how the effects of tau treatment in APOE3-Ch astrocytes differ from those in WT astrocytes, relative to each genotype's no-treatment condition. This interaction analysis revealed that, upon tau treatment, APOE3-Ch astrocytes upregulate genes involved in cell projection assembly, while WT astrocytes showed downregulation in this process, along with concurrent upregulation of cell cycle–related genes. Astrocyte cell projections, or extensions, are critical for cellular communication[29] and for the clearance of toxic oligomers such as tau.[30]

Close examination (Figure 5F) revealed that several key genes involved in regulating cell assembly of cell projections, including *TENM1*, *TBC1D8 TBC1D30*, *STAU2*, *PLD1*, *DYNLT2B*, *DAAM2*, *CDC42EP4*, and *ADAMTS16*, were downregulated in tau-treated WT astrocytes. In contrast, these genes were upregulated in APOE3-Ch astrocytes after tau treatment, suggesting that APOE3-Ch astrocytes enhance their ability to form projections, possibly to aid in the clearance of tau oligomers.

In parallel, genes associated with the endocytosis pathway, such as *RIN1*, *RAB7B*, *PLD2*, *MYD88*, *MFGE8*, *MAPKAPK3*, *LETMD1*, *EEF2K*, *CHRNA7*, *CD81*, and *ANK2*, also followed a similar pattern (Figure 5G). These genes are involved in all aspects of endocytosis. For instance, *RIN1* and *PLD2* have been implicated in activating receptor-mediated endocytosis,[31, 32] *RAB7B* in late endosomes,[33, 34] and *CD81* in exosomes and multivesicular bodies (MVBs).[35] APOE3-Ch astrocytes upregulated these genes upon tau treatment, suggesting an

F, Schematic of the tau clearance assay with inhibitors of the protein degradation pathways, including Bafilomycin A1 and MG132. G, Percentage of tau remaining after treatment with the lysosomal inhibitor Bafilomycin A1 (BafA1) or vehicle (0.5 μM, 16 hours), measured by flow cytometry in *APOE ε3* or APOE3-Ch astrocytes treated with Tau-AF647 (100 nM, 3 hours, $n = 6$, 10,000 cells each). Statistical significance is determined by two-way ANOVA with Šídák multiple comparison tests. H, Percentage of tau remaining after treatment with the proteasomal inhibitor MG132 or vehicle (15 μM, 16 hours), measured by flow cytometry in *APOE ε3* or APOE3-Ch astrocytes treated with tau-AF647 (100 nM, 3 hours, $n = 3$, 10,000 cells each). Statistical significance is determined by two-way ANOVA with Šídák multiple comparison tests. All data are expressed as mean ± standard deviation with individual data points shown. The results designated as "ns" are not significant; $^{*}P < 0.05$; $^{**}P < 0.01$; $^{***}P < 0.001$; $^{****}P < 0.0001$. ANOVA, analysis of variance; APOE3-Ch, APOE3 Christchurch; RM, repeated measures; WT, wild type

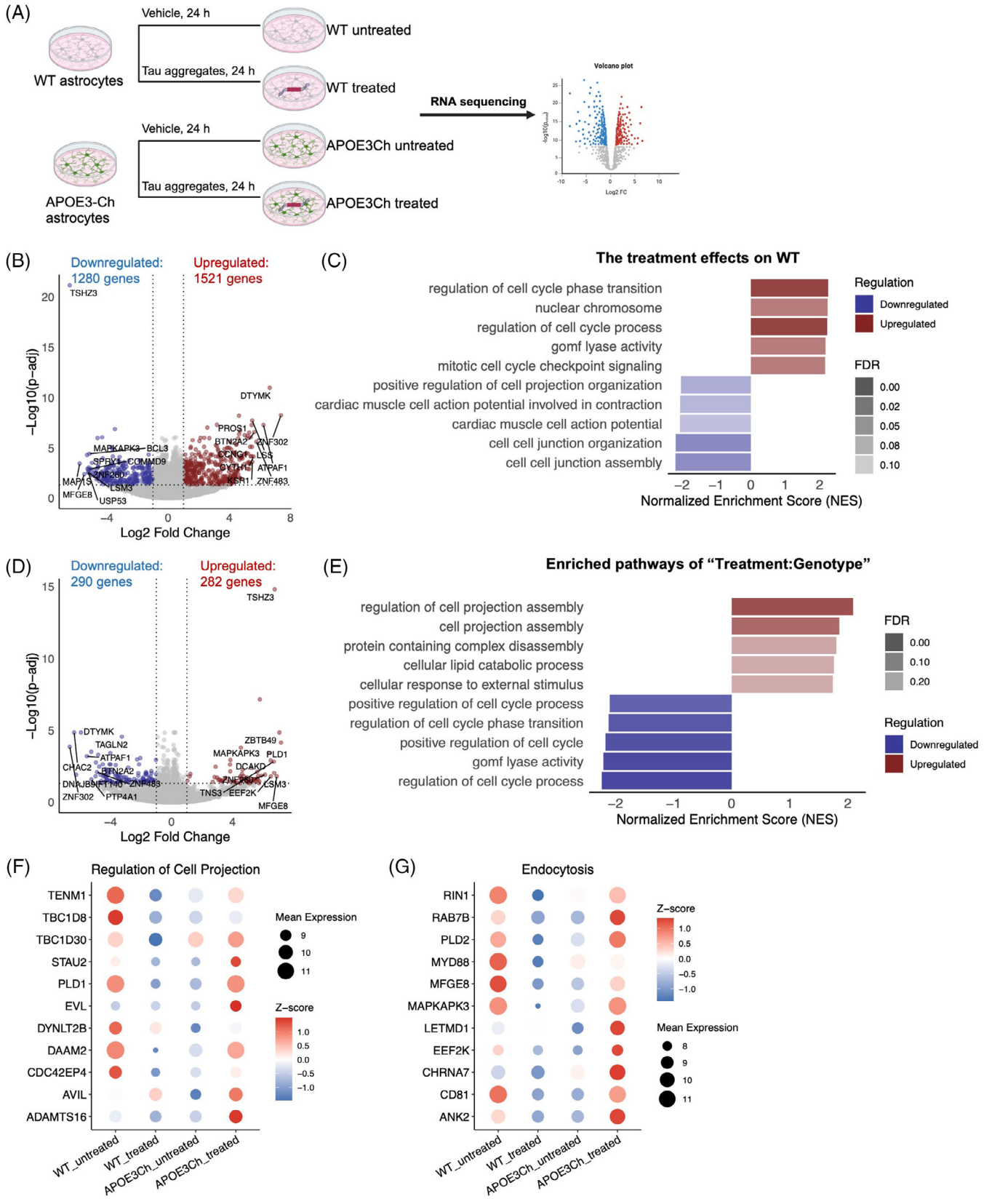


FIGURE 5 Transcriptomics studies reveal APOE3-Ch astrocytes upregulate genes related to cell projection and endocytosis upon tau treatment. A, Schematics of the samples processed by RNA sequencing. B, Volcano plot showing differentially expressed (DE) genes in WT-treated astrocytes compared to WT-untreated astrocytes. Statistically significant (*Padj* < 0.05, |Log2 fold change| > 1) DE genes are labeled in red (upregulated) and blue (downregulated). The top 10 upregulated and downregulated genes ranked by their fold change are labeled with their corresponding gene symbols. C, Gene set enrichment analysis (GSEA) results of the top upregulated pathways (red) and downregulated pathways

enhancement of endocytosis and subsequent lysosomal degradation pathways. This aligns with our prior findings that APOE3-Ch astrocytes show enhanced tau internalization and degradation through the lysosomal pathway.

In summary, our transcriptomic analysis reveals a complex and coordinated defense mechanism in APOE3-Ch astrocytes in response to tau pathology. This involves upregulation of cell projections and endocytosis pathways, facilitating enhanced uptake and degradation of tau, which may confer protective effects against tau toxicity.

4 | DISCUSSION

In this study, we characterized the phenotypic and functional differences between WT and APOE3-Ch astrocytes, uncovering several key differences in lipid metabolism, tau uptake, and clearance mechanisms. We found that APOE3-Ch astrocytes exhibit lipid dyshomeostasis, characterized by decreased lipid buffering capacity and impaired apoE secretion. Lipidomic analysis revealed significantly reduced ceramide levels in APOE3-Ch astrocytes, which is likely driven by the upregulation of ceramide catabolic pathways. Our analysis of tau uptake demonstrated that APOE3-Ch astrocytes internalize tau oligomers more efficiently than WT astrocytes, a process mediated by both HSPG and LRP1 pathways. After tau internalization, APOE3-Ch astrocytes degraded tau through both lysosomal and proteasomal pathways, while WT astrocytes primarily relied on proteasomal degradation and secreted more tau than APOE3-Ch astrocytes. This decrease in secreted tau in APOE3-Ch astrocytes may contribute to the limited tau spread in the human case expressing the Christchurch variant.[10] Finally, transcriptomic analysis showed upregulation of cell projection and endocytic pathways in APOE3-Ch astrocytes, the latter potentially enhancing their ability to internalize and degrade tau oligomers. These transcriptional changes align with our cellular assay findings and suggest that APOE3-Ch astrocytes have a heightened capacity for clearing tau.

APOE3-Ch is a rare genetic variant of *APOE* and has been associated with HLPP3 in patients.[21] APOE3-Ch mice have exhibited peripheral dyslipidemia, including elevated *APOE* and *APOB* levels, as well as increased association of *APOE* with VLDL and low-density lipoproteins LDL.[8] However, the effects of APOE3-Ch on CNS lipid metabolism remain unclear. Our findings suggest that similar to astrocytes expressing *APOE* ε4—an *APOE* variant known to increase risk for AD and cardiovascular diseases—APOE3-Ch astrocytes show reduced capacity to buffer lipids, as evidenced by increased lipid droplet accumulation after OA stimulation. One possible explanation for this accumulation is impaired lipid secretion, supported by the observed reduction in apoE levels in the media. However, this remains speculative, and further studies are needed to directly test this mechanism.[36] Our results differ from those of Murakami et al.,[37] who used similarly differentiated iPSC-derived APOE3-Ch astrocytes but reported no genotype-dependent differences in lipid droplet accumulation or apoE secretion. These discrepancies may stem from technical differences. In our lipid loading assay, we used a moderate dose of 20 μM OA, whereas Murakami et al. used a higher dose of 200 μM. The higher OA concentration may have saturated lipid storage pathways, masking subtle genotype-specific effects. For apoE secretion, we used a commercially validated ELISA kit with high sensitivity, while Murakami et al. used a custom assay that may be more variable. Differences in assay sensitivity and detection thresholds could account for the divergent findings. We also acknowledge the possibility of biological variability between cell lines.

Unlike *APOE* ε4 astrocytes, APOE3-Ch astrocytes do not accumulate unsaturated TGs, which are prone to lipid peroxidation.[38] The accumulation of unsaturated TGs in *APOE* ε4 astrocytes may promote oxidative stress and pro-inflammatory profiles, contributing to neuroinflammation and neurotoxicity,[38] processes that are not observed in APOE3-Ch astrocytes.

Additionally, our data show that APOE3-Ch astrocytes have reduced levels of ceramides both intracellularly and in the media. Ceramides are bioactive lipids that play a significant role in several pathological processes associated with AD, including Aβ plaque formation, mitochondrial dysfunction, cellular senescence, and autophagy dysfunction.[24] Ceramides are also known to initiate some of the pathological changes that drive AD progression.[24, 25] The lower ceramide levels observed in APOE3-Ch astrocytes may explain, in part, why this variant is protective against AD, despite the observed lipid dyshomeostasis. Although we observed a modest, non-significant increase in lipid droplet formation in APOE3-Ch astrocytes under physiological conditions, Cers are predominantly localized to cellular membranes.[39] As such, any Cers stored in lipid droplets—such as acylceramides—may not substantially contribute to the total ceramide levels detected by whole-cell lipidomics, suggesting that the observed decrease in specific Cer levels likely reflects alterations in membrane-associated Cer metabolism.

(blue) ranked by their normalized enrichment score (NES) comparing WT-treated astrocytes to WT-untreated astrocytes. Pathways with a false discovery rate (FDR) < 0.1 are shown and the FDR of each pathway is represented on a color transparency scale. D, Volcano plot showing DE genes in the interaction term, "Treatment:Genotype." Statistically significant (*P*adj < 0.05, |Log2 fold change| > 1) DE genes are labeled in red (upregulated) and blue (downregulated). The top 10 upregulated and downregulated genes ranked by their fold change are labeled with their corresponding gene symbols. E, GSEA results of the top upregulated pathways (red) and downregulated pathways (blue) ranked by their NES in the interaction term, "Treatment:Genotype." Pathways with a FDR < 0.3 are shown and the FDR of each pathway is represented on a color transparency scale. F, Quantification of the mean expression and *z* score of each DEG (*P*.adj < 0.05, abs(Log2FC) ≥ 1) involved in the GO Biological Process "Regulation of Cell Projection." The size of the dot represents the mean expression, and the *z* score is plotted on a color gradient. G, Quantification of the mean expression and z-score of each DEG (*P*.adj < 0.05, abs(Log2FC) ≥ 1) involved in the GO Biological Process "Endocytosis." The size of the dot represents the mean expression, and the *z* score is plotted on a color gradient. APOE3-Ch, APOE3 Christchurch; GO, Gene Ontology; WT, wild type

The role of astrocytic *APOE* in tau-mediated neurodegeneration has been highlighted by recent work showing that selective removal of astrocytic *APOE* ε4 strongly protects against tau-induced synaptic loss and reduces microglial-mediated synaptic phagocytosis.[40] This underscores the isoform-specific effects of *APOE* on tau pathology and neurodegeneration. While *APOE* ε4 impairs astrocytic function and exacerbates tau toxicity, our findings suggest that APOE3-Ch astrocytes confer protection by enhancing tau clearance via HSPG- and LRP1-mediated pathways. Previous studies have shown that astrocytes can internalize tau monomers,[13] truncated pre-formed fibrils (PFFs),[12] and tau oligomers[11,41] through HSPG-independent mechanisms.[28] Our data suggest that, like neurons, which rely heavily on LRP1-mediated and HSPG-mediated uptake,[20,42,43] APOE3-Ch astrocytes use both HSPG- and LRP1-mediated pathways for tau oligomer internalization, while WT astrocytes primarily rely on LRP1-mediated tau uptake. The dual mechanisms of LRP1- and HSPG-mediated uptake in APOE3-Ch astrocytes are reminiscent of their synergistic roles in Aβ uptake observed in primary neurons.[44] Research indicates that LRP1 can form complexes with HSPGs, allowing ligands such as *APOE* to either bind directly to the LRP1–HSPG complex or initially interact with HSPGs before being transferred to LRP1 for endocytosis.[45] However, our study does not address whether HSPGs and LRP1 cooperate directly in tau uptake in APOE3-Ch astrocytes or elucidate the involvement of specific endocytosis pathways, such as macropinocytosis or receptor-mediated endocytosis. The dual engagement of these two mechanisms represents a novel protective feature of the APOE3-Ch variant, allowing for more efficient tau uptake and clearance from the extracellular space.

After tau internalization, our data show that APOE3-Ch astrocytes degrade tau primarily through lysosomal and proteasomal pathways, with lysosomal degradation playing a particularly vital role. In contrast, WT astrocytes rely more heavily on proteasomal degradation and secrete a greater proportion of tau. Lysosomal degradation in APOE3-Ch astrocytes occurs more rapidly, leading to enhanced tau clearance and preventing the accumulation of tau oligomers. The transcriptomic upregulation of endocytic and cell projection pathways in APOE3-Ch astrocytes aligns with these findings, as these pathways likely enhance the cells' ability to internalize and degrade tau. By scavenging tau oligomers more effectively, APOE3-Ch astrocytes may prevent the spread of tau pathology and mitigate neurotoxicity.

Our findings complement recent work by Chen et al.,[8] who investigated APOE3-Ch in a humanized APOE3-Ch knock-in mouse model. Notably, their analysis of APOE3-Ch myeloid cells demonstrated significantly enhanced uptake of tau fibrils—mediated by both HSPGs and LRP1—along with increased degradation and reduced tau release after phagocytosis, suggesting a cell-autonomous, protective role for APOE3-Ch in limiting tau spread. While their study focused primarily on microglial mechanisms in the presence of amyloid pathology, our data reveal that APOE3-Ch astrocytes exhibit altered tau handling and lipid metabolism under amyloid-independent conditions. Specifically, we observed enhanced tau internalization and intracellular degradation via lysosomal and proteasomal pathways, along with reduced extracellular tau accumulation. These findings align with the mechanisms reported in APOE3-Ch myeloid cells and suggest that APOE3-Ch promotes tau clearance through parallel, yet cell type–specific, mechanisms. Collectively, these studies support a model in which APOE3-Ch confers neuroprotection through coordinated, multicellular actions that enhance tau degradation and reduce extracellular tau propagation.

Our findings contribute to the growing understanding of APOE3-Ch as a protective variant against AD. Unlike the *APOE* ε4 variant, which has been associated with both impaired lipid metabolism and tau-induced neurotoxicity, APOE3-Ch astrocytes may offer protection through two mechanisms: enhanced tau clearance and reduced harmful lipid levels. The enhanced tau uptake and degradation observed in APOE3-Ch astrocytes may help to prevent tau accumulation and spreading, a hallmark of AD progression. The reduced levels of ceramides and GLAs observed in APOE3-Ch astrocytes may confer protection by dampening bioactive lipid-mediated signaling pathways known to exacerbate AD pathology. However, the functional implications of this reduction remain to be fully clarified. One possibility is that lower ceramide levels reflect decreased pro-apoptotic signaling or reduced cellular stress responses, thereby contributing to neuroprotection. Alternatively, the elevated ceramide levels commonly observed in AD could represent a compensatory response or a byproduct of disease progression rather than a primary pathogenic driver. Further studies are needed to determine whether reduced ceramide levels in APOE3-Ch astrocytes are causally protective or simply a reflection of altered lipid homeostasis.

Although our study highlights these protective mechanisms, the observed lipid dysregulation in APOE3-Ch astrocytes suggests that this variant's effects are complex. While the variant helps clear tau and reduce neurotoxic lipids, it may also impair normal lipid metabolism, potentially leading to long-term vulnerabilities in astrocyte–neuron metabolic coupling. Future studies should aim to better understand how the lipid dysregulation we observed may influence astrocyte function over time and whether it poses risks for other neurodegenerative conditions.

A limitation of our study is the use of iPSC-derived astrocytes, which may not fully replicate the mature astrocyte phenotype observed in vivo. For example, the low expression of *AQP4* in our astrocytes may suggest incomplete maturation, which could affect the generalizability of our findings. Future research should focus on validating these results in more mature astrocyte models or in vivo systems to ensure that the observed phenotypes hold true in fully developed astrocytes.

Additionally, further investigation is needed to understand the mechanisms underlying the reduced exported *APOE* observed in APOE3-Ch astrocytes. Given the central role of *APOE* in lipid transport and metabolism, understanding how this reduction impacts lipid handling and its broader implications for brain health could provide new insights into the role of *APOE* variants in neurodegenerative diseases. Finally, future studies should explore the effects of APOE3-Ch astrocytes on neuronal function, particularly in co-culture or animal models, to better understand how these cells influence tauopathy and other neurodegenerative processes.

AUTHOR CONTRIBUTIONS

Xinran Tian and Kenneth S. Kosik designed research; Xinran Tian, Erica Keane Rivera, Arina A. Nikitina, Inyoung Hwang, Jennifer A. Smith, Youngsuh J. Cho, Julia Sala-Jarque, Austin DuBose, Clare Andriola, Toby Gollan-Myers performed research; Xinran Tian and Kenneth S. Kosik analyzed data; and Xinran Tian and Kenneth S. Kosik wrote the article.

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CONFLICT OF INTEREST STATEMENT

Kenneth S. Kosik consults for Expansion Therapeutics and is a member of the Rainwater Charitable Foundation Board of Directors. Kenneth S. Kosik owns stock options from Minerva Biotechnologies Corp and ADRx. The remaining authors declare no competing interests. Author disclosures are available in the supporting information.

CONSENT STATEMENT

This study does not involve human subjects; therefore, informed consent is not required.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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